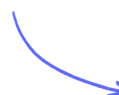


Types of Graphs

Quadratic Graphs / Cubic Graphs / Reciprocal Graphs / Exponential Graphs

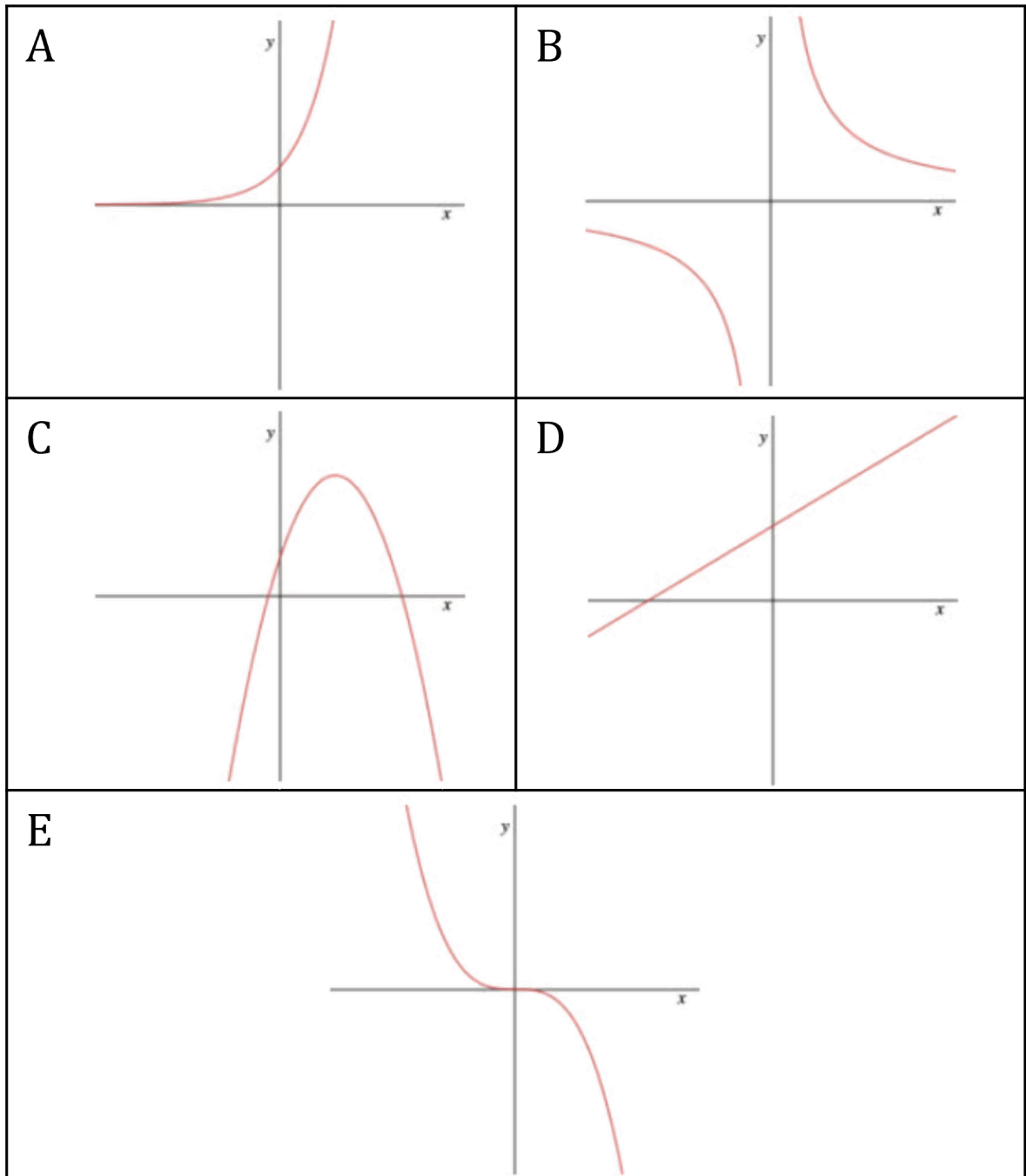
Easy (3 questions)	/6
Medium (5 questions)	/11
Hard (3 questions)	/11
Total Marks	/28

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Easy Questions

1 Match the graphs to the equations.



$$(1) y = \frac{1}{2}x + 3, \quad (2) y = 2^x, \quad (3) y = -2.1x^3,$$

(4) $y = \frac{3}{x}$, (5) $y = -2x^2 + 3x + 1$

(3 marks)

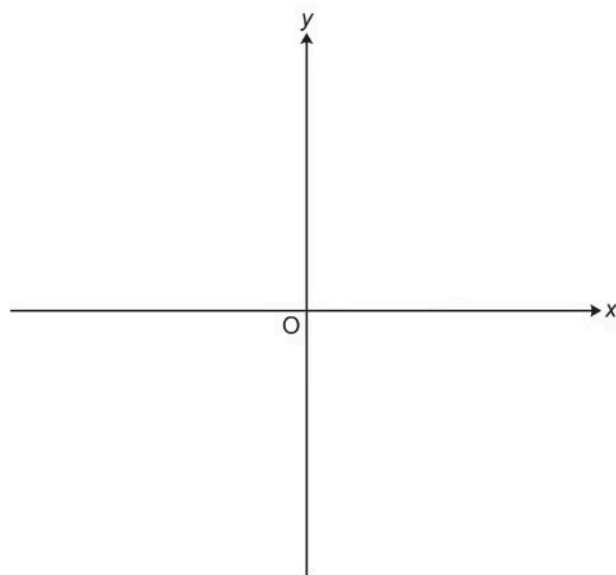
- 2 The equation of a curve is $y = (x-1)^2 - 6$

Write down the coordinates of the turning point.

(1 mark)

- 3 Sketch the graph of $y = 3^x$

Give the value of the y -intercept.



(2 marks)

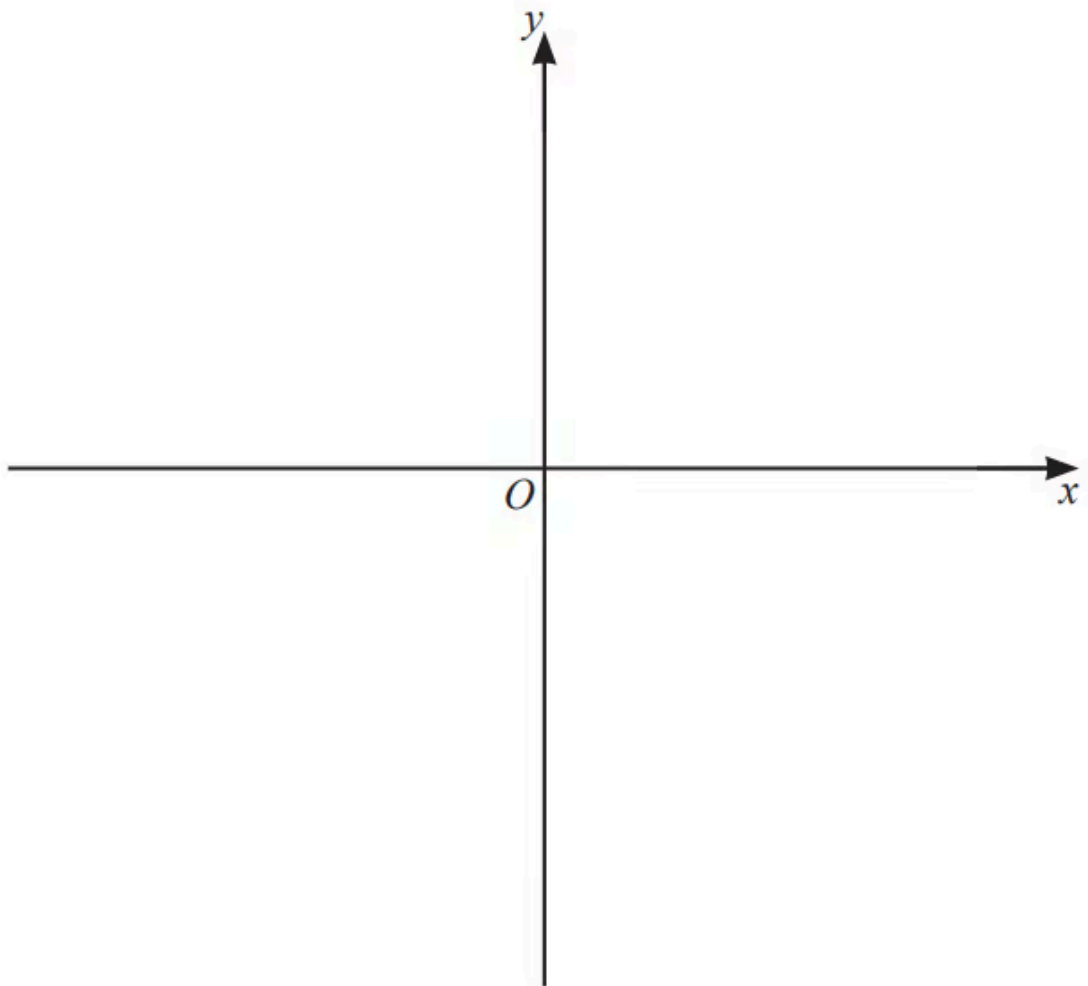
Medium Questions

- 1 Write down the equation of the line of symmetry of the graph with equation $y = (x - 2)(x + 2)$.

(1 mark)

- 2 On the diagram, sketch the graph of $y = (x + 1)(x - 3)^2$.

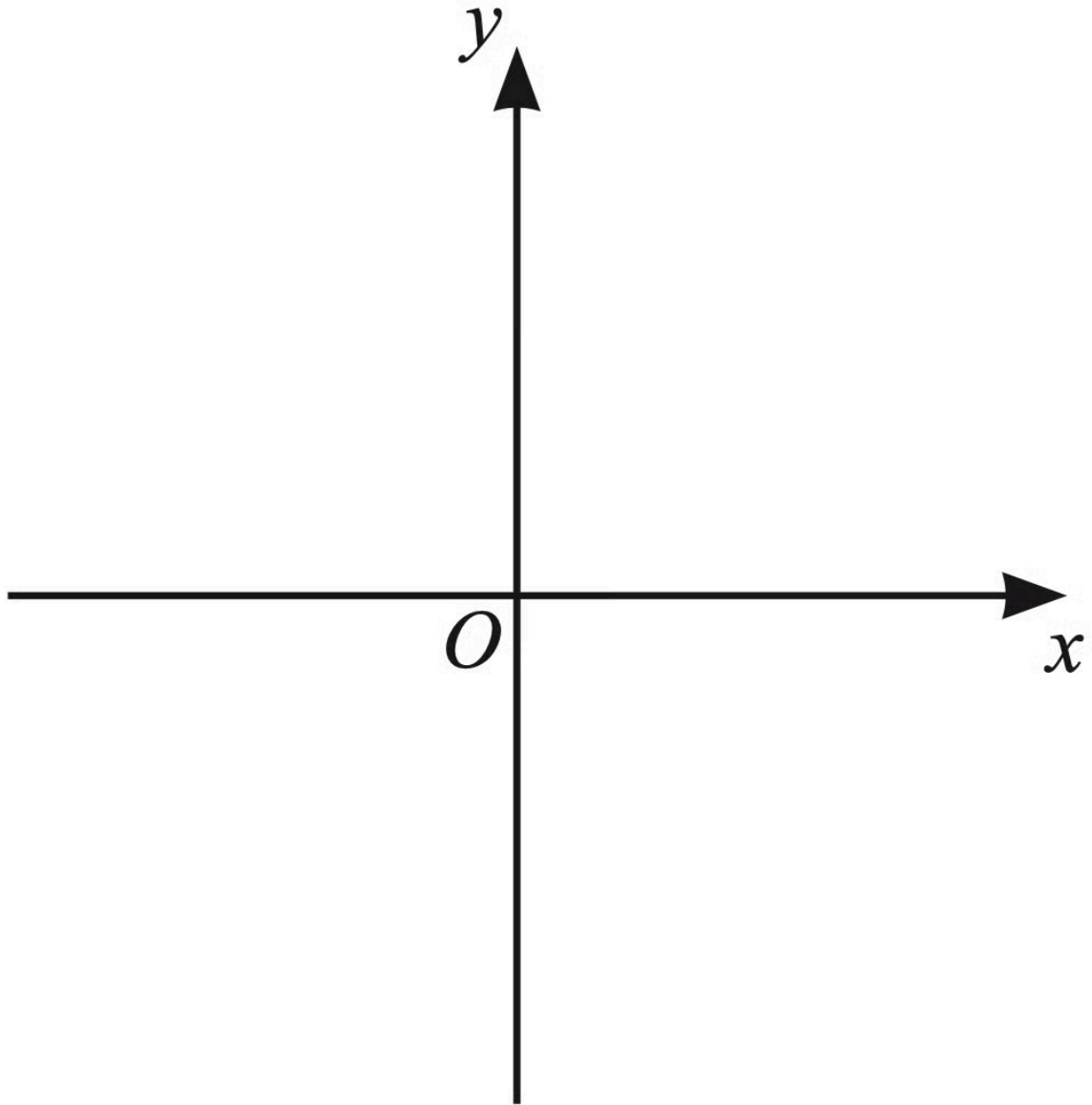
Label the values where the graph meets the x -axis and the y -axis.



(4 marks)

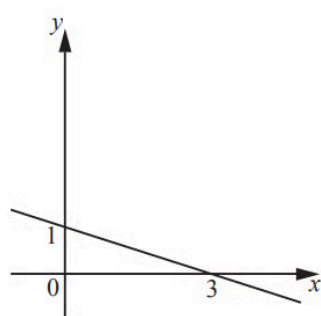
3 Sketch the graph of the function.

$$y = \frac{1}{x}$$

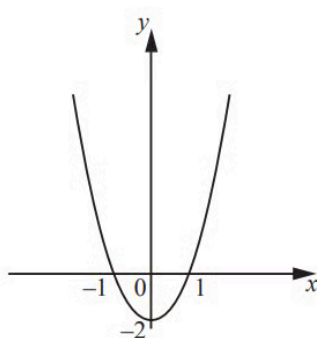


(2 marks)

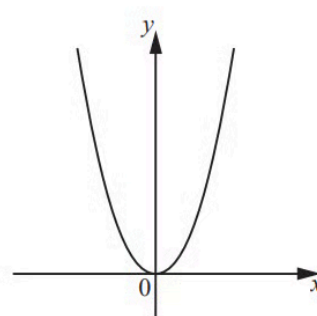
4 The diagrams A, B, C, D, E and F are six graphs of different functions.



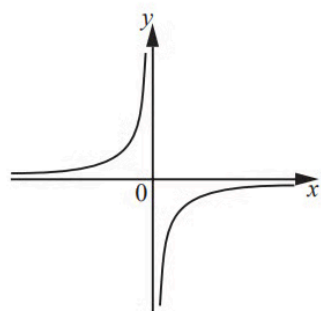
A



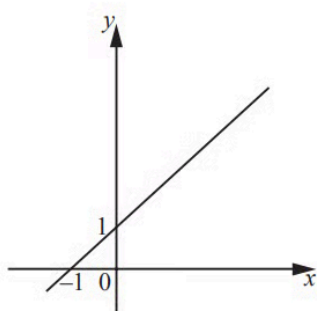
B



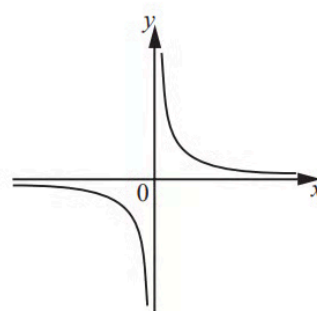
C



D



E



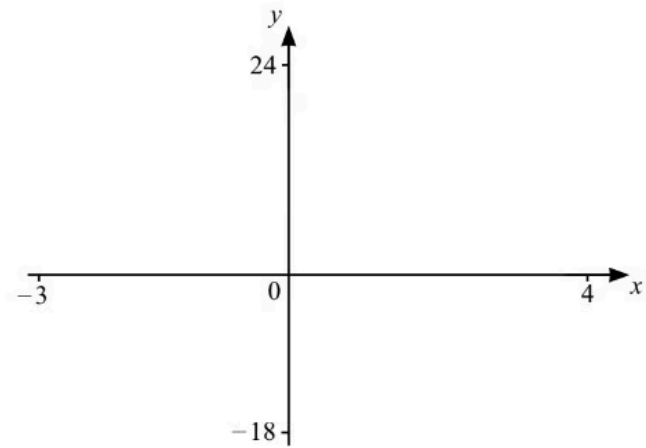
F

Complete the table to identify the correct graph for each function. One has been done for you.

Function	$y = x + 1$	$y = 1 - \frac{x}{3}$	$y = 2x^2$	$y = -\frac{4}{x}$
Diagram	E			

(3 marks)

5



$$f(x) = x(x+2)(x-3)$$

On the diagram, sketch the graph of $y = f(x)$ for $-3 \leq x \leq 4$.
Show the values of the intersections with the axes.

(1 mark)

Hard Questions

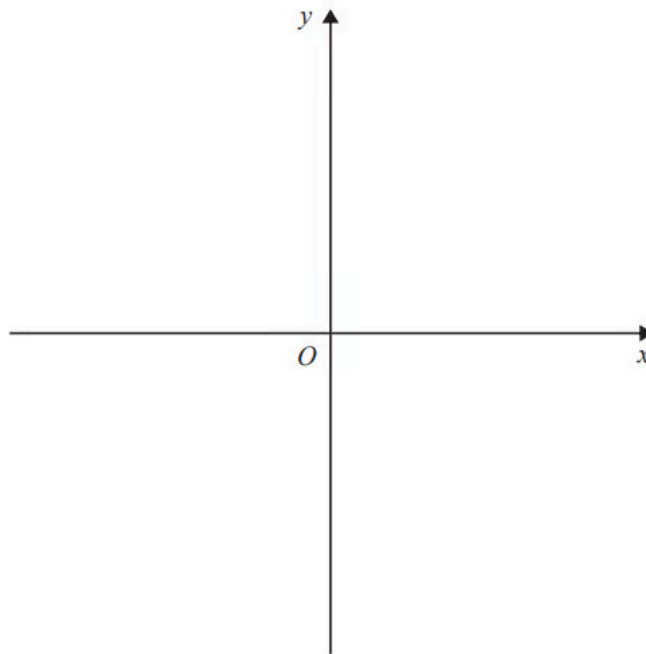
- 1 The curve C has equation $y = (x - 3)^2 - 5$

Using the axes below, sketch the curve C .

On your sketch show clearly

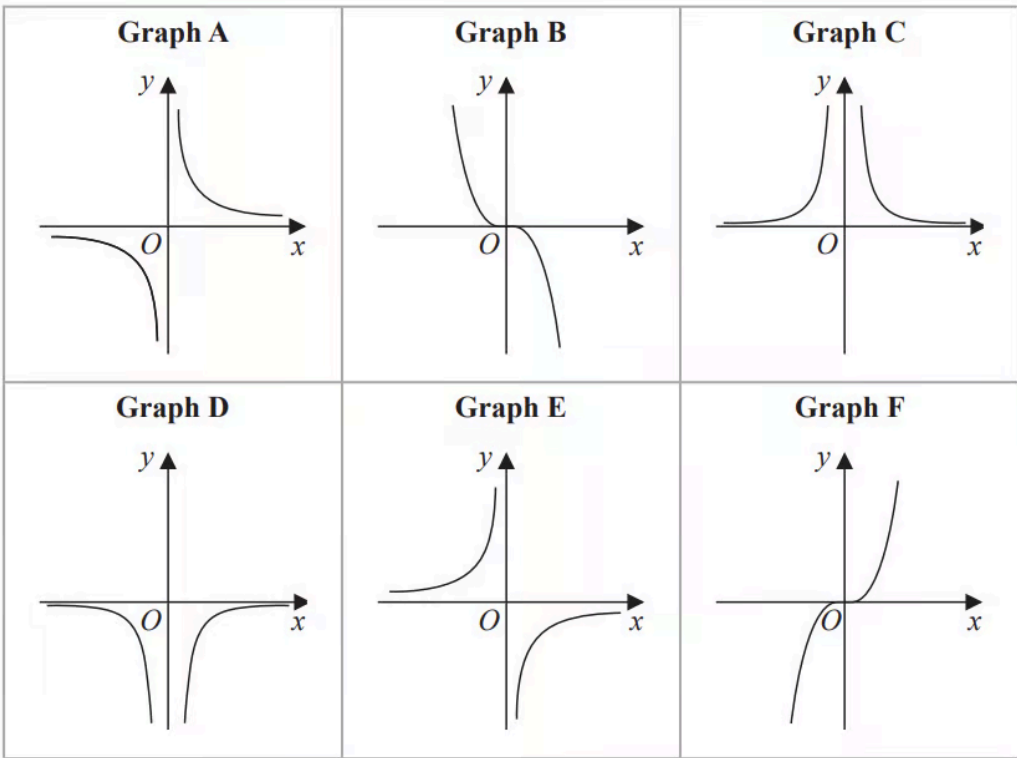
(i) the exact coordinates of any points of intersection of C with the coordinate axes,

(ii) the coordinates of the turning point.



(4 marks)

2 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	
$y = -\frac{1}{2}x^3$	
$y = -\frac{5}{x}$	

(3 marks)

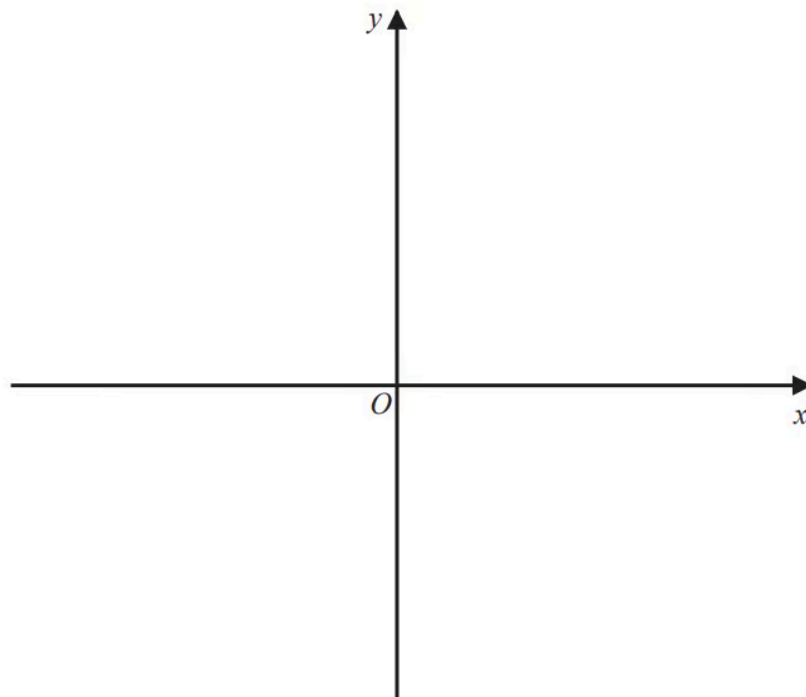
- 3 The curve C has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve C .

On your sketch show clearly, in terms of a ,

(i) the coordinates of any points of intersection of C with the coordinate axes,

(ii) the coordinates of the turning point.



(4 marks)